

Document 03 — Dashboard Design, Modes & Metrics

The product

How the analysis becomes something a reviewer can operate: the layout, the two data modes, the map's color logic, the ranked readouts, Investigator Mode, and exactly which metric appears where and what each one means.

Assumes: Documents 01 (the problem) and 02 (what the numbers are and how they are built).

Subject: the evolved v2 dashboard — a single self-contained HTML page, no build step, no server-side code.

Prepared: July 2026 · public-data prototype.

1. Design principles

The interface follows a small set of rules that keep it honest and legible under the same constraints as the analysis.

Map-left, rail-right

The map stays visible at all times (no bottom-sheet covering it); the right rail is a real fixed column of readouts and controls, not a hidden drawer.

One visual system

A single diverging color scale means the same thing everywhere — **red** is "runs empty" (deficit), **blue** is "runs full" (surplus), in both data modes.

Precomputed, never live-computed

Every control reads a precomputed JSON file. Nothing re-runs a model in the browser — the page is a viewer over reproducible outputs, which is what lets it run on a \$0 static host.

Graceful degradation

Any optional layer whose data file fails to load simply does not appear, rather than breaking the page. If the historical data is missing entirely, a clearly-labeled illustrative demo layer stands in, with an on-screen banner saying so.

2. The two data modes

A single top-bar toggle switches the entire map between two sources. They answer different questions and use different underlying quantities, so the legend re-labels itself when you switch:

Mode	Underlying metric	Answers	Legend reads
Historical flow (default)	Net daily flow (bikes/day) for the selected hour	Where does imbalance <i>chronically</i> concentrate?	Net daily flow · -2.7 ... 0 ... +2.7
Live docks	Dock fill % (bikes ÷ capacity) from GBFS	What is the <i>actual current</i> fill state?	Dock fill · 0% ... 100%

The two are genuinely different quantities (a modeled per-day rate vs. a real-time percentage), so the tool never blends them into one number — it re-labels the scale instead.

3. The map and its color logic

Stations are drawn as markers colored by a **5-class diverging ramp** (ColorBrewer RdBu-5): deep red → light red → near-neutral → light blue → deep blue. Discrete classes, not a continuous gradient, so the legend can match the map exactly and small markers stay readable.

THE SCALE IS A REAL COMPUTED STATISTIC, NOT A ROUND NUMBER

The endpoints are $\pm$ the 95th percentile of |net flow| across every station, hour, and period — currently ≈ 2.74 **bikes/day** (shown as ± 2.7). Anchoring on the p95 (with a small floor) means the deepest colors mark genuinely unusual imbalance, without a handful of extreme outliers washing everything else pale.

Two refinements make the map readable at NYC's station density:

- **The "balanced" class recedes.** Near-zero stations are drawn faint, with a soft gray stroke rather than white, so they fade into the background instead of competing with the stations that actually need attention — and, critically, so they do not vanish white-on-white against the pale basemap.
- **View follows zoom.** At a city-wide zoom the ~2,500 markers would over-plot into a solid mass, so the map starts in **Grouped** view (clustered counts) and auto-switches to **Individual** markers at street zoom ($z \geq 14$). A manual Individual/Grouped toggle overrides the auto-switch when the user wants control. Selecting a station grows its marker and darkens its ring so the colored fill stays visible rather than being swamped by the highlight.

4. Time and slice controls

- **24-hour time slider** with a play button that animates through the day (paused by default, and it respects reduced-motion preferences).
- **Weekday / Weekend** toggle.
- **Period selector** — all-period average, each season, or each individual analysis month — built directly from the data's own coverage, so it never offers a slice the data cannot support.

The slider sits in a compact bar centered along the bottom of the map, on the same row as the legend, so the bottom edge reads as one control strip.

5. The permanent readout — headline numbers always in view

Above the interactive controls, four tiles keep the population-level picture visible without any interaction. At the default settings they read (all reproduced from the committed data):

Tile	Value	Meaning
AM deficit stations	1,037	Stations flagged short of bikes across the 6–9 AM commute window
Near NYCHA / school	1,585 / 2,516	Within 300 m of a NYCHA development or school (default threshold)
Beyond subway gap	357	More than 800 m from the nearest subway entrance (default threshold)
Current map filter	–	Echoes the active hour / day-type / period so a screenshot is self-describing

Equity tiles recompute live as the Investigator Mode thresholds move — they are not baked-in constants.

6. "Worst now" — two ranked, clickable lists

The stations under the most pressure right now, surfaced as the keyboard- and screen-reader-accessible equivalent of the map (there is otherwise no keyboard path to 2,516 SVG circles). Split into two lists so each is internally comparable:

- **Most bikes needed** — the 8 largest **deficits** (drawn red).
- **Most docks needed** — the 8 largest **surpluses** (drawn blue).

They were deliberately split from a single magnitude-ranked list: a full-dock station is not "worst" in the same sense as an empty one, and mixing them hid which problem a station actually had. Clicking an entry flies the map to that station and opens its detail panel.

7. Investigator Mode — the what-if sandbox

The tool's differentiator, open by default so a first-time viewer sees it. A running **diff bar** summarizes baseline-vs-scenario for all three controls at once — zero-centered bars encoding the *change*, not the absolute level — and stays visible even when the panel is collapsed, so an active non-default scenario is never lost. Below it, four collapsible sections:

Control	What it does	Reads
Equity thresholds	Two sliders (NYCHA/school proximity, subway gap) recompute the flagged counts live from each station's real distances	<code>flows.json</code> context
Fleet simulator	A 1–10-truck slider swaps in a precomputed scenario and redraws the real route on the map	<code>fleet_scenarios.json</code>
Weather scenario	Temperature/precipitation sliders (or a preset) project each station's flow through its own elasticity; the map recolors live	<code>elasticities.json</code>
Load a shared scenario	Paste a shared link to re-apply a saved scenario	URL state

Weather presets: *Ideal* (22.2 °C), *Steady rain* (15.6 °C / 10.2 mm), *Snow event* (−2.2 °C / 7.6 mm), *Hot day* (31.0 °C). Every value is inside the real observed range.

HONESTY BUILT INTO THE CONTROL ITSELF

The weather panel states in place that its projection is *estimated from historical elasticity, not a weather-specific model*, and names the 18-zone 6 km grid it reads from. The fleet panel states that added trucks do not show diminishing returns at this scale because each truck hits its 45-stop shift cap. These are not footnotes — they sit next to the sliders.

8. Station detail

Selecting any station (on the map or from a ranked list) opens a detail card: its name and typology role, a 24-hour **flow-rhythm strip** (the same signed curve the map colors, drawn small), and its **per-station equity context** — nearest NYCHA development, school, and subway entrance, each with a real distance. This is where per-station equity lives, by design (Document 01, §6.2). A "More info" disclosure holds the cluster typology and the school-layer vintage note.

9. Model performance panel

A small, quiet disclosure near the footer holds the real walk-forward result from Document 02 — the three-tier MAE table showing the seasonal-naïve baseline winning. It is intentionally low-emphasis (a static reference, read once) but it is *present*: the tool shows the reviewer the evidence that its own forecast is the simple one, rather than asserting sophistication it did not earn.

10. Metric reference — what appears where

Metric	Where	Unit / scale	Source
Net daily flow	Map markers (historical), detail strip, legend	bikes/day, ± 2.7 (p95)	<code>flows.json</code>
Dock fill %	Map markers (live)	0–100%	GBFS live
AM-deficit count	Readout tile	count (1,037)	<code>route.json</code>
Equity proximity counts	Readout tiles; equity sliders	counts (live)	<code>flows.json</code> context
Worst-now rankings	Two ranked lists	top 8 each, bikes/day	<code>flows.json</code>
Fleet serviced	Fleet panel; diff bar	stations serviced	<code>fleet_scenarios.json</code>
Weather projection	Weather panel; map recolor	% shift in magnitude	<code>elasticities.json</code>
Forecast MAE (3 tiers)	Model performance panel	bikes/day (1.95 / 2.18 / 2.10)	<code>model_performance.json</code>

11. Accessibility & responsiveness

- **WCAG AA color contrast** — muted text colors were audited against their real backgrounds and darkened until every one clears 4.5:1.
- **Real keyboard operation** — the segmented mode/view/day toggles are true ARIA radiogroups with arrow-key navigation (not just roles), and the ranked lists give a keyboard path to any station. The hour slider exposes a formatted `aria-valuetext` ("8:00 AM", not "8").
- **Touch targets & motion** — controls grow to 44 px on touch pointers; animation is disabled under `prefers-reduced-motion`.
- **Mobile layout** — the map becomes sticky so it stays visible while the controls scroll, the masthead compacts, and Investigator Mode is promoted to the top of the mobile rail so the differentiator is not buried.

NOTE ON DESIGN ITERATION

The v2 design described here is being refined in a parallel dashboard file so the original stays intact while changes are validated (headless-browser behavior checks and screenshots per change). Which file becomes the single shipped dashboard is a decision reserved for the end of the project — the design principles and metric contract in this document apply to whichever one ships.

